CAGE-Fake: Article Trustworthiness Audit Report
Calibration-Aware Gating & Evidence-Grounded Reasoning for Trustworthy Fake News Triage

AUDIT ID: AUD-198561
DATE: 20261028_201202
FRAMEWORK: CAGE-Fake v2.0

LEGAL & FACT-CHECKING POLICY: A trained fake-news classifier predicts linguistic, stylometric, and semantic patterns learned from benchmark datasets; it does **NOT** independently verify factual reality. URL extraction, pattern recognition, and external evidence verification remain strictly separate processes. The pipeline abstains (**UNCERTAIN**) whenever evidence is missing, conflicting, or model consensus is low.

Title: Man gets life term for abducting, raping minor
Source: https://timesofindia.indiatimes.com/city/agra/man-gets-life-te...
Word Count: 215 words | **Extraction:** Successfully Extracted via Trafilatura
Triage Policy: Evidence-Grounded Three-Way Gating

TRIAGE VERDICT: FAKE
Calibrated Confidence: 94.9% (τ = 80.0%)
Model Disagreement (std): 0.217 (δ = 0.25)
Safety Trigger: Tri Gate Accepted

1. Multi-Model Disagreement Analysis (5 Models)

Model Name	P(FAKE)	Vote	Confidence
Random Forest	42.4%	REAL	57.6%
XGBoost	95.0%	FAKE	95.0%
LightGBM	95.1%	FAKE	95.1%
Logistic Regression	71.1%	FAKE	71.1%
CAGE-Fake Stack	94.9%	FAKE	94.9%

Consensus Status: Moderate Agreement
Vote Distribution: 4 FAKE vs 1 REAL
Base Disagreement (4-base std): 0.217 (δ = 0.25)
5-Model Consensus Spread: 0.208
[==========------]
Safety Assessment: Minor divergence observed, but within acceptable uncertainty limits (std <= delta).

2. News Credibility Fingerprint (Linguistic & Stylistic Scorecard)

Credibility Dimension	Score (0-100)	Journalistic Assessment
Sensationalism Discipline	90.3/100	Excellent / Restrained
Source Attribution	74.4/100	Adequate / Neutral
Epistemic Grounding	90.3/100	Excellent / Restrained
Punctuation Discipline	88.2/100	Excellent / Restrained
Readability Structure	90.3/100	Excellent / Restrained
Lexical Diversity	78.5/100	Adequate / Neutral

OVERALL CREDIBILITY INDEX
84.9 / 100
Credibility Tier: A (Restrained / Professional Copy)

Scores are standardized against verified newsroom copy standards: high factual attribution, balanced syntax, and low sensationalism maximize the index.

3. Grounded Diagnostic Reasoning & Counterfactual Sensitivity

Executive Synthesis: Predictive models identify deceptive stylometric patterns and emotional sensationalism. Triaged as FAKE (94.9% confidence).

Diagnostic Explanation: 1. Decision & Triage Status: The CAGE-Fake pipeline reached a verdict of **FAKE** with **94.9% calibrated confidence**.
- **Model Consensus**: Moderate Agreement (4/5 models voted FAKE, average P(FAKE)=79.7%).
- **Driving Deceptive Markers**: LR [TF-IDF Lexical], XGB [SBERT Semantic], LightGBM [SVD Topics].
- **Credibility Score**: Article credibility index is 84.9/100 (A (Restrained / Professional Copy)).

Counterfactual Simulation: Neutralizing 1 linguistic markers resulted in a signed probability shift of -3.2 percentage points (from 94.9% to 91.7%), retaining the primary verdict (FAKE). Original P(FAKE): **94.9%** → Transformed P(FAKE): **91.7%** **(Signed Shift: -3.2%)**

[v] Clean Stylometrics: No significant sensational exaggeration, clickbait patterns, or manipulative phrasing detected.

4. Claim-Level Semantic Evidence Grounding (Triage Verification)

Extracted Factual Claim	Verification Status	Authoritative Source / Match	Evidence Context Quote
Agra: A special Pocso court in Agra has sentenced a 30-year-old man to life impris...	SUPPORTED (Sim: 0.96)	timesofindia.indiatimes.com	Agra: A special Pocso court in UP's Kasganj has sentenced a 30-year-old man to 20 ...
During the trial, the prosecution examined 12 witnesses, including the complainant...	SUPPORTED (Sim: 0.86)	medicaldialogues.in	“During the trial, the court examined 12 prosecution witnesses, of whom six, inclu...
While delivering the verdict, special judge (Pocso) Sonika Chaudhary highlighted s...	SUPPORTED (Sim: 1.05)	timesofindia.indiatimes.com	While delivering the verdict, special judge (Pocso) Sonika Chaudhary highlighted s...
Man gets life term for abducting, raping minor.	SUPPORTED (Sim: 0.76)	abc3340.com	# Life behind bars for man convicted of kidnapping and raping 10-year-old The Blo...

5. Local Signal Attributions (Meta-Ensemble Impacts)

Top FAKE-Driving Indicators	Impact	Top REAL-Driving Anchors	Impact
LR [TF-IDF Lexical]	+1.562	RF [Stylometrics]	-0.213
XGB [SBERT Semantic]	+0.876	Inter-View Disagreement (std)	-0.202
LightGBM [SVD Topics]	+0.630	-	-

6. Empirical Benchmark Model Comparison (Held-out Test Split)

Model Name	Accuracy	Macro-F1	ECE	Coverage	Selective Acc
Random Forest (Uncalibrated)	0.8098	0.8097	0.0770	100.0%	0.8098
XGBoost (Uncalibrated)	0.8508	0.8505	0.0550	100.0%	0.8508
LightGBM (Uncalibrated)	0.8787	0.8785	0.0401	100.0%	0.8787
Logistic Regression (Uncalibrated)	0.9361	0.9361	0.1322	100.0%	0.9361
Uncalibrated Stack	0.9410	0.9410	0.0182	100.0%	0.9410
Calibrated Stack (Full Coverage)	0.9410	0.9409	0.0274	100.0%	0.9410
CAGE-Fake (Calibrated + Gated)	**0.9410**	**0.9409**	**0.0274**	**82.5%**	**0.9781**

7. Research Calibration & Disagreement Visualizations







